PBL Portal Content - Aryan Daware

Smart Hospital Management System (SmartCare) | Team Nobita (PBL2627-AI&DS-A-031;) | Mentor: Er. Ram Babu Buri

Member	Aryan Daware (24E1ARADM30P020)
Module	Module 4 - Pharmacy, Laboratory and Nursing
Code owned	Medicine, MedicineInventory, Prescription, PrescriptionItem, LabTest, PatientVital models and DAOs, MedicineServlet (/inventory), PrescriptionServlet, LabServlet, NurseServlet, inventory, prescription_view, lab_management, lab_test_details, nurse JSPs.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

How to use this file

1) Daily Work Log tab: pick the Date, type Hours Worked, keep Status as shown, paste the description, click Submit Today's Work. 2) Weekly Report tab: choose the Week, paste the GitHub link, then paste the three boxes (Work Completed, Problems Faced, Plan for Next Week) from Part A. 3) Part A (weekly) comes first, Part B has the day-wise table. Submit your own entries only, from your own login.

PART A - Weekly Reports (13 weeks)

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Joined team of five and brainstormed hospital system ideas. (2) Studied pharmacy and laboratory handling in hospitals. (3) Wrote the abstract section on prescriptions, lab tests and vitals. (4) Reviewed guide's expectations on specialisation. (5) Reviewed the abstract and helped with submission.
Problems Faced	Too many ideas to compare. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Listed requirements for medicine stock and expiry tracking. (2) Listed requirements for doctor prescriptions with medicine items. (3) Listed requirements for lab test request and result entry.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Listed requirements for medicine stock and expiry tracking. (2) Listed requirements for doctor prescriptions with medicine items. (3) Listed requirements for lab test request and result entry. (4) Listed requirements for nurse vitals recording. (5) Combined these into the pharmacy, lab and nursing SRS section.
Problems Faced	Many pharmacy rules to capture. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for Nurse and Lab actors. (2) Designed the class diagram for Medicine and MedicineInventory. (3) Designed the class diagram for Prescription and PrescriptionItem.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Drew the use-case diagram for Nurse and Lab actors. (2) Designed the class diagram for Medicine and MedicineInventory. (3) Designed the class diagram for Prescription and PrescriptionItem. (4) Designed the class diagram for LabTest and PatientVital. (5) Drew the prescription sequence diagram and merged UML.
Problems Faced	Parent-child relation for prescription items. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed the medicines and medicine_inventory tables. (2) Designed prescriptions and prescription_items tables. (3) Designed lab_tests and patient_vitals tables.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Designed the medicines and medicine_inventory tables. (2) Designed prescriptions and prescription_items tables. (3) Designed lab_tests and patient_vitals tables. (4) Set up JDK 17, Tomcat and MySQL locally and cloned the repo. (5) Drew mock-ups for inventory, prescription and vitals screens.
Problems Faced	Foreign key order in schema. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote Medicine and MedicineInventory model classes. (2) Wrote Prescription and PrescriptionItem models. (3) Wrote LabTest and PatientVital models.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote Medicine and MedicineInventory model classes. (2) Wrote Prescription and PrescriptionItem models. (3) Wrote LabTest and PatientVital models. (4) Implemented MedicineDAO with stock queries. (5) Implemented PrescriptionDAO, LabDAO and VitalDAO inserts.
Problems Faced	Mapping decimal fields correctly. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Implemented MedicineServlet for inventory add and update. (2) Implemented PrescriptionServlet to save multiple medicine items. (3) Implemented LabServlet for test request and result update.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Implemented MedicineServlet for inventory add and update. (2) Implemented PrescriptionServlet to save multiple medicine items. (3) Implemented LabServlet for test request and result update. (4) Implemented NurseServlet for recording and viewing vitals. (5) Added low stock and expiry checks in the DAO layer.
Problems Faced	Saving many items needed a loop with transaction. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created the inventory.jsp with stock table and add form. (2) Created prescriptions.jsp for the doctor and prescription_view.jsp. (3) Created lab_management.jsp and lab_test_details.jsp.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Created the inventory.jsp with stock table and add form. (2) Created prescriptions.jsp for the doctor and prescription_view.jsp. (3) Created lab_management.jsp and lab_test_details.jsp. (4) Created the nurse dashboard.jsp and patients.jsp. (5) Created nurse record_vitals.jsp and vitals_view.jsp.
Problems Faced	Dynamic rows in prescription form. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Added validation for quantity, dose and vitals ranges. (2) Handled exceptions for stock below zero cases. (3) Added JavaScript validation to vitals form.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Added validation for quantity, dose and vitals ranges. (2) Handled exceptions for stock below zero cases. (3) Added JavaScript validation to vitals form. (4) Restricted lab and inventory pages by role. (5) Tested validation messages on every form.
Problems Faced	Stock could go negative. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Reduced stock automatically when a prescription is issued. (2) Added lab result status flow (Requested to Completed). (3) Added vital history list for each patient.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Reduced stock automatically when a prescription is issued. (2) Added lab result status flow (Requested to Completed). (3) Added vital history list for each patient. (4) Reviewed code, renamed methods and added comments. (5) Pushed module code to GitHub and updated the sheet.
Problems Faced	Lab status flow had gaps. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Linked prescriptions with consultation from Module 3. (2) Linked lab tests to the doctor's lab_reports page. (3) Linked patient prescriptions page with prescription data.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Linked prescriptions with consultation from Module 3. (2) Linked lab tests to the doctor's lab_reports page. (3) Linked patient prescriptions page with prescription data. (4) Fixed foreign key errors after merging with other modules. (5) Ran end-to-end flow from consultation to prescription and lab test.
Problems Faced	Foreign key errors after merge. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote test cases for inventory and prescription. (2) Tested stock deduction with multiple medicines. (3) Tested lab request and result entry.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote test cases for inventory and prescription. (2) Tested stock deduction with multiple medicines. (3) Tested lab request and result entry. (4) Tested nurse vitals for valid and invalid ranges. (5) Fixed issues and retested all test cases.
Problems Faced	Edge case in stock deduction. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Checked inventory and prescription pages on the live site. (2) Fixed foreign key insert order in seed data. (3) Corrected date display in the lab details page.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Checked inventory and prescription pages on the live site. (2) Fixed foreign key insert order in seed data. (3) Corrected date display in the lab details page. (4) Retested nurse vitals on the live server. (5) Noted deployment fixes for the report.
Problems Faced	Seed data failed on cloud DB. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the pharmacy, lab and nursing section of the report. (2) Added screenshots of inventory, prescription and vitals. (3) Prepared slides on pharmacy and lab workflow.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote the pharmacy, lab and nursing section of the report. (2) Added screenshots of inventory, prescription and vitals. (3) Prepared slides on pharmacy and lab workflow. (4) Rehearsed demo of prescription and lab result. (5) Prepared viva answers on transactions and foreign keys.
Problems Faced	Many screenshots to collect. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

PART B - Daily Work Log (Monday to Friday, 65 entries)

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description
06-07-2026 (Mon)	1.75	Completed	Joined team of five and brainstormed hospital system ideas.
07-07-2026 (Tue)	1.5	Completed	Studied pharmacy and laboratory handling in hospitals.
08-07-2026 (Wed)	1.5	Completed	Wrote the abstract section on prescriptions, lab tests and vitals.
09-07-2026 (Thu)	1.75	Completed	Reviewed guide's expectations on specialisation.
10-07-2026 (Fri)	1.75	Completed	Reviewed the abstract and helped with submission.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description
13-07-2026 (Mon)	2	Completed	Listed requirements for medicine stock and expiry tracking.
14-07-2026 (Tue)	1.75	Completed	Listed requirements for doctor prescriptions with medicine items.
15-07-2026 (Wed)	1.5	Completed	Listed requirements for lab test request and result entry.
16-07-2026 (Thu)	2	Completed	Listed requirements for nurse vitals recording.
17-07-2026 (Fri)	2	Completed	Combined these into the pharmacy, lab and nursing SRS section.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description
20-07-2026 (Mon)	1.75	Completed	Drew the use-case diagram for Nurse and Lab actors.
21-07-2026 (Tue)	1.75	Completed	Designed the class diagram for Medicine and MedicineInventory.
22-07-2026 (Wed)	1.75	Completed	Designed the class diagram for Prescription and PrescriptionItem.
23-07-2026 (Thu)	1.75	Completed	Designed the class diagram for LabTest and PatientVital.
24-07-2026 (Fri)	1.75	Completed	Drew the prescription sequence diagram and merged UML.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description
27-07-2026 (Mon)	1.5	Completed	Designed the medicines and medicine_inventory tables.
28-07-2026 (Tue)	1.75	Completed	Designed prescriptions and prescription_items tables.
29-07-2026 (Wed)	1.75	Completed	Designed lab_tests and patient_vitals tables.
30-07-2026 (Thu)	1.75	Completed	Set up JDK 17, Tomcat and MySQL locally and cloned the repo.
31-07-2026 (Fri)	2	Completed	Drew mock-ups for inventory, prescription and vitals screens.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description
03-08-2026 (Mon)	1.75	Completed	Wrote Medicine and MedicineInventory model classes.
04-08-2026 (Tue)	1.75	Completed	Wrote Prescription and PrescriptionItem models.
05-08-2026 (Wed)	1.5	Completed	Wrote LabTest and PatientVital models.
06-08-2026 (Thu)	1.75	Completed	Implemented MedicineDAO with stock queries.
07-08-2026 (Fri)	1.5	Completed	Implemented PrescriptionDAO, LabDAO and VitalDAO inserts.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description
10-08-2026 (Mon)	1.75	Completed	Implemented MedicineServlet for inventory add and update.
11-08-2026 (Tue)	1.75	Completed	Implemented PrescriptionServlet to save multiple medicine items.
12-08-2026 (Wed)	2	Completed	Implemented LabServlet for test request and result update.
13-08-2026 (Thu)	1.5	Completed	Implemented NurseServlet for recording and viewing vitals.
14-08-2026 (Fri)	1.75	Completed	Added low stock and expiry checks in the DAO layer.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description
17-08-2026 (Mon)	1.75	Completed	Created the inventory.jsp with stock table and add form.
18-08-2026 (Tue)	1.5	Completed	Created prescriptions.jsp for the doctor and prescription_view.jsp.
19-08-2026 (Wed)	2	Completed	Created lab_management.jsp and lab_test_details.jsp.
20-08-2026 (Thu)	1.75	Completed	Created the nurse dashboard.jsp and patients.jsp.
21-08-2026 (Fri)	1.75	Completed	Created nurse record_vitals.jsp and vitals_view.jsp.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description
24-08-2026 (Mon)	1.75	Completed	Added validation for quantity, dose and vitals ranges.
25-08-2026 (Tue)	1.5	Completed	Handled exceptions for stock below zero cases.
26-08-2026 (Wed)	1.5	Completed	Added JavaScript validation to vitals form.
27-08-2026 (Thu)	1.5	Completed	Restricted lab and inventory pages by role.
28-08-2026 (Fri)	1.75	Completed	Tested validation messages on every form.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description
31-08-2026 (Mon)	2	Completed	Reduced stock automatically when a prescription is issued.
01-09-2026 (Tue)	1.75	Completed	Added lab result status flow (Requested to Completed).
02-09-2026 (Wed)	2	Completed	Added vital history list for each patient.
03-09-2026 (Thu)	1.5	Completed	Reviewed code, renamed methods and added comments.
04-09-2026 (Fri)	1.5	Completed	Pushed module code to GitHub and updated the sheet.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description
07-09-2026 (Mon)	1.5	Completed	Linked prescriptions with consultation from Module 3.
08-09-2026 (Tue)	1.5	Completed	Linked lab tests to the doctor's lab_reports page.
09-09-2026 (Wed)	1.5	Completed	Linked patient prescriptions page with prescription data.
10-09-2026 (Thu)	1.75	Completed	Fixed foreign key errors after merging with other modules.
11-09-2026 (Fri)	1.75	Completed	Ran end-to-end flow from consultation to prescription and lab test.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description
14-09-2026 (Mon)	1.75	Completed	Wrote test cases for inventory and prescription.
15-09-2026 (Tue)	2	Completed	Tested stock deduction with multiple medicines.
16-09-2026 (Wed)	1.75	Completed	Tested lab request and result entry.
17-09-2026 (Thu)	1.75	Completed	Tested nurse vitals for valid and invalid ranges.
18-09-2026 (Fri)	1.75	Completed	Fixed issues and retested all test cases.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description
21-09-2026 (Mon)	2	Completed	Checked inventory and prescription pages on the live site.
22-09-2026 (Tue)	1.5	Completed	Fixed foreign key insert order in seed data.
23-09-2026 (Wed)	1.5	Completed	Corrected date display in the lab details page.
24-09-2026 (Thu)	2	Completed	Retested nurse vitals on the live server.
25-09-2026 (Fri)	1.5	Completed	Noted deployment fixes for the report.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description
28-09-2026 (Mon)	2	Completed	Wrote the pharmacy, lab and nursing section of the report.
29-09-2026 (Tue)	1.5	Completed	Added screenshots of inventory, prescription and vitals.
30-09-2026 (Wed)	2	Completed	Prepared slides on pharmacy and lab workflow.
01-10-2026 (Thu)	1.5	Completed	Rehearsed demo of prescription and lab result.
02-10-2026 (Fri)	2	Completed	Prepared viva answers on transactions and foreign keys.

Total hours in this log: 112 over 65 days.